

UNIVERSITY OF PISA
DEPARTMENT OF PHYSICS
MASTER'S DEGREE IN PHYSICS OF COMPLEX SYSTEMS

Anomalous Transport in Soaring Flights

Collective Strategies and Intermittent Search Models

Candidate

Matteo Di Sante

Supervisors

Prof. Michael Benzaquen

Dr Alexandre Darmon

Internal tutor (Pisa)

Prof. Riccardo Mannella

Academic year 2025–2026

Working draft — generated on 2026-06-28.

Sources and code: <https://github.com/matteodisante/soaring-anomalous-transport>

Abstract

Soaring birds and human pilots stay aloft by extracting energy from atmospheric updrafts, producing long, intermittent trajectories that alternate climbs within thermals with quasi-ballistic glides. Such motion is a candidate for *anomalous transport*, that is, a departure from ordinary Brownian diffusion. This thesis characterizes the transport statistics of real soaring flights from a large empirical dataset and compares them with stochastic transport models.

This document reports the current state of the project. A first, reproducible acquisition has collected 203,343 flights spanning 27 seasons (1999–2000 to 2025–2026) from the FFVL Coupe Fédérale de Distance, of which 186,225 carry a GPS tracklog and 186,052 have been downloaded and verified. Data collection is ongoing and may be extended to further public soaring databases (for example XContest), using the same scraping approach. The remainder of the document describes the acquisition method, the dataset and its statistics, and the planned analysis and modelling.

Contents

1	Introduction	2
1.1	Motivation	2
1.2	Objectives	2
1.3	Scope of this document	3
2	Data acquisition	4
2.1	Source: the FFVL Coupe Fédérale de Distance	4
2.2	What is collected	4
2.3	Pipeline and tooling	4
2.4	Reproducibility	5
2.5	Integrity	5
2.6	Further data sources	5
3	The dataset	6
3.1	The IGC tracklog format	6
3.2	The catalog	6
3.3	Coverage and statistics	6
3.4	Data quality and caveats	7
4	Next steps	9
4.1	Trajectory pre-processing	9
4.2	Observables on the whole trajectory	9
4.3	Segmentation into flight phases	10
4.4	Phase-resolved observables	11
4.5	Reproducing the reference results	12
4.6	Transport analysis	12
4.7	Modelling and simulation	12
4.8	Tooling	13

Chapter 1

Introduction

1.1 Motivation

Gliders, paragliders, and soaring birds carry no continuous source of propulsion; to remain airborne they extract energy from the moving atmosphere, primarily from *thermals*, the buoyant updrafts that form as sunlight heats the ground. A cross-country flight thus proceeds as a repeating cycle of three phases. In the *transition* phase the flyer glides along a nearly straight, energy-efficient path toward the next source of lift. If none is reached at a safe altitude, a *search* phase follows, during which the flyer probes the surrounding air mass to locate lift. Once lift is found, a *climb* phase begins, in which the flyer circles within the rising air to regain altitude. The cycle alternates long, directed displacements with episodes of slow, localized motion.

Intermittent and directionally persistent motion of this kind is the setting in which *anomalous transport* arises. Let $r(t)$ denote the horizontal position of a flight. Transport is anomalous when the mean-squared displacement grows as a non-trivial power of time, $\langle r^2(t) \rangle \sim t^\alpha$ with $\alpha \neq 1$, as opposed to the linear growth ($\alpha = 1$) of ordinary Brownian diffusion. Continuous-time random walks and Lévy walks are the standard frameworks for such regimes [1, 2], which have been reported in animal movement, in disordered media, and in turbulent flows. Whether the trajectories of real soaring flights belong to an anomalous regime, and of which type, is the question addressed in this thesis.

Soaring has previously been studied mainly as a control problem—how to centre and exploit a thermal efficiently, including by reinforcement learning [3]. The present work is statistical instead: it asks which transport properties emerge at the level of the ensemble of observed trajectories. Its direct precedent is the study of Vilpellet et al. [4], who analysed a large set of human soaring flights, found a horizontal transport that is sub-ballistic and approximately independent of aircraft type (Hurst exponent $H \approx 0.88$), and introduced a phase-resolved analysis based on the transition–search–climb cycle. This thesis builds on that study: it enlarges the dataset and re-examines both the phase segmentation and the transport analysis.

1.2 Objectives

The thesis pursues three objectives:

1. to assemble a large, reproducible, and documented dataset of real soaring flights;
2. to reconstruct clean trajectories from the raw GPS logs and define suitable transport observables;

3. to test for anomalous transport and to compare the empirical statistics with stochastic transport models.

A first dataset—the FFVL CFD archive—has been assembled and verified and is described here; data collection is ongoing and may incorporate further sources (Chapter 2). The second and third objectives define the work ahead (Chapter 4).

1.3 Scope of this document

This document reports the current state of the project and is revised as the work proceeds. Its quantitative content—the summary statistics and tables—is generated directly from the dataset, so it remains consistent with the data in hand. Chapter 2 describes the acquisition of the data and the reproducibility of the pipeline; Chapter 3 describes the dataset and its statistics; and Chapter 4 sets out the planned analysis and modelling.

Chapter 2

Data acquisition

2.1 Source: the FFVL Coupe Fédérale de Distance

The data come from the *Coupe Fédérale de Distance* (CFD), the cross-country competition run by the French free-flight federation (FFVL). Pilots upload their flights; each entry carries metadata (pilot, date, take-off and landing sites, scored distance, duration, glider) and, in most cases, the raw GPS tracklog in IGC format. The competition publishes one public list per season, addressable as a machine-readable XML export. A “season” is identified by its starting year, e.g. the year 1999 denotes the 1999–2000 season.

2.2 What is collected

For every season the pipeline retrieves two things:

- the **season XML export**, archived verbatim as provenance (`raw_xml/<year>.xml`); it is the authoritative list of flights and links;
- the **IGC tracklog files** referenced by the XML, one file per flight that has a track.

From these, a derived **catalog** is built (`catalog.csv`, one row per flight, metadata plus the local path of its track) together with a per-season index (`seasons_index.csv`). The raw data (tens of gigabytes) live on an external disk and are never committed to the repository; only the small per-season index is versioned, and it is the source of the statistics in Chapter 3.

2.3 Pipeline and tooling

Acquisition is performed by a dedicated, installable Python package (`soaring.acquisition.ffvl`) exposed through a command-line tool, `soaring-ffvl`, with five subcommands: `fetch-xml` (archive the season XMLs), `download` (fetch the IGC files), `build-catalog` (regenerate the catalog and the index), `status` (per-season coverage), and `verify` (on-disk integrity check). The design favours robustness for a long-running, large job:

Resumable downloads. Files already present are skipped, so the job can be interrupted and resumed freely.

Atomic writes. Each file is written to a temporary `.part` and then renamed, so a file under its final name is always complete – important on the exFAT external disk, which has no journaling.

Content validation. A response is accepted only if it looks like a real IGC file (an **A** header record followed by at least one **B** fix record); HTML error pages and truncated responses are rejected and retried.

Courteous networking. A small bounded thread pool, per-request timeouts, exponential backoff, and a minimum delay between requests avoid hammering the server; a browser TLS fingerprint is impersonated to pass the site’s Cloudflare challenge.

2.4 Reproducibility

The destination directory (`data_root`) is not hard-coded: it is set per machine through an environment variable (`SOARING_FFVL_DATA_ROOT`) or a configuration file shipping a placeholder, so no machine-specific path is committed. Every command validates this setting at start-up and aborts with an explicit message if the target is missing, rather than failing obscurely. Archiving the season XMLs verbatim keeps the full provenance: the catalog can always be rebuilt from them with `build-catalog`.

2.5 Integrity

Because the data sit on an external drive, the `verify` subcommand re-scans every downloaded file on disk, reusing the same IGC validation applied at download time, and flags truncated or invalid files, leftover `.part` fragments, and read errors. The FFVL collection has been verified in full: all 186,052 downloaded tracklogs pass validation (Chapter 3).

2.6 Further data sources

The FFVL CFD is the first and, at present, the primary source, but the dataset is not assumed to be final. Additional public soaring repositories are being pursued—notably XContest, and possibly others—to enlarge and diversify the sample. These repositories expose comparable flight metadata and IGC tracks, so they can be acquired with the same scraping approach and merged into the same catalog. The analysis is designed to treat the resulting collection uniformly, stratifying by source where this matters.

Chapter 3

The dataset

3.1 The IGC tracklog format

A flight track is stored as an IGC file, the standard plain-text format of the Fédération Aéronautique Internationale for flight recorders. Each line is a typed record identified by its first letter. The relevant ones are:

- A** the logger/manufacturer identifier (first record);
- H** header records (date of flight, pilot, glider, datum, ...);
- I** the declaration of optional fields appended to each fix;
- B** the GPS fixes – the core of the file. A B record carries a UTC timestamp, latitude, longitude, a validity flag, and both the pressure (barometric) and GPS altitudes.

The sequence of B records is therefore a time-stamped 3-D trajectory. The sampling period is typically a few seconds but is not guaranteed to be uniform and varies with the recording device – a point the analysis stage must handle explicitly.

3.2 The catalog

Two derived tables make the raw files usable. The **catalog** (`catalog.csv`) has one row per flight: the competition metadata plus a `local_path` column linking the flight to its file on disk, so a flight can be traced to its track (and back, via the flight identifier embedded in the file name) without any external lookup. The **season index** (`seasons_index.csv`), versioned in the repository, summarizes each season with the number of flights declared, the number carrying a GPS track, and the number actually downloaded.

3.3 Coverage and statistics

The statistics below describe the FFVL CFD dataset currently in hand; further sources may be added later (Chapter 2). The dataset spans 27 seasons, from 1999–2000 to 2025–2026. It contains 203,343 flights in total; 186,225 of them (91.6 %) carry a GPS tracklog, and of those 186,052 (99.9 %) have been downloaded and verified. The per-season breakdown is given in Table 3.1. The volume of flights grows by more than three orders of magnitude over the period, reflecting the adoption of GPS loggers and online scoring: the early seasons contribute a handful of tracks each, while recent seasons contribute well over ten thousand.

Table 3.1: FFVL CFD flights per season: total declared, with a GPS track, and downloaded. Source: `data/seasons_index.csv`.

Season	Flights	With GPS	Downloaded	Coverage (%)
1999–2000	10	10	10	100.0
2000–2001	832	1	1	100.0
2001–2002	1,150	3	3	100.0
2002–2003	1,996	123	123	100.0
2003–2004	1,470	337	337	100.0
2004–2005	1,765	574	574	100.0
2005–2006	2,162	894	894	100.0
2006–2007	2,386	1,016	1,016	100.0
2007–2008	3,081	1,667	1,667	100.0
2008–2009	3,974	2,614	2,613	100.0
2009–2010	4,171	3,059	3,059	100.0
2010–2011	4,872	4,185	4,169	99.6
2011–2012	6,744	6,075	6,065	99.8
2012–2013	7,873	7,106	7,075	99.6
2013–2014	9,804	9,157	9,127	99.7
2014–2015	13,793	13,293	13,254	99.7
2015–2016	13,982	13,703	13,696	99.9
2016–2017	15,760	15,534	15,528	100.0
2017–2018	14,917	14,682	14,676	100.0
2018–2019	16,702	16,517	16,508	99.9
2019–2020	10,966	10,868	10,863	100.0
2020–2021	9,500	9,483	9,480	100.0
2021–2022	14,552	14,507	14,502	100.0
2022–2023	11,590	11,561	11,560	100.0
2023–2024	10,372	10,362	10,360	100.0
2024–2025	10,906	10,885	10,884	100.0
2025–2026	8,013	8,009	8,008	100.0
Total	203,343	186,225	186,052	99.9

3.4 Data quality and caveats

A few limitations are already known and will shape the analysis:

- **Flights without a track.** About 8 % of entries have no IGC file (older or manually declared flights); they carry metadata only.
- **A small residue of unrecoverable files.** A few flights advertise a track that the server does not actually serve as a valid IGC (broken or placeholder links); these are recorded as failures and excluded.
- **Heterogeneous logging.** Sampling rate, altitude source, and accuracy depend on the device; trajectories must be resampled and quality-filtered before any ensemble statistic is computed.
- **Incomplete dates.** Some historical entries have placeholder dates (e.g. 0000–00–00); these are kept but flagged.

These caveats motivate the pre-processing described in Chapter 4.

Chapter 4

Next steps

With the dataset in place and verified, the work proceeds as follows.

4.1 Trajectory pre-processing

Parse the IGC B records into clean trajectories of (time, latitude, longitude, altitude); project the geographic coordinates to a local metric frame; resample to a common cadence to absorb device heterogeneity; and apply quality filters (drop fixes with bad validity flags, implausible speeds, or large gaps). The output is a tidy, analysis-ready representation, produced by a future `soaring.analysis` sub-package. Throughout, $r(t)$ denotes the horizontal position and $v(t) = \dot{r}(t)$ the horizontal velocity.

4.2 Observables on the whole trajectory

A first group of observables requires *no* segmentation: it can be measured directly on the full trajectories and already probes the transport at the global level. For each, we state what it is and what it is for.

Transport and diffusion.

Mean-squared displacement (MSD). The average squared horizontal displacement over a time lag t , $\text{MSD}(t) = \langle |r(t_0 + t) - r(t_0)|^2 \rangle$, averaged over start times t_0 and over flights. Its growth law $\text{MSD}(t) \sim t^\alpha$ defines the transport regime ($\alpha = 1$ Brownian, $\alpha > 1$ super-diffusive, $\alpha < 1$ sub-diffusive; equivalently the Hurst exponent $H = \alpha/2$). This is the primary test for anomalous transport.

Time- versus ensemble-averaged MSD. The MSD computed as a time average along a single flight, compared with the average across flights. A systematic gap between the two – with the time average depending on the observation window – signals *ergodicity breaking* and aging, hallmarks of trapping (CTRW-type) dynamics, and thus helps discriminate between candidate models.

Displacement distribution $P(\Delta r, t)$. The probability distribution of the horizontal displacement $\Delta r = r(t_0 + t) - r(t_0)$ at a fixed lag t (the propagator of the walk). Its shape separates ordinary diffusion (Gaussian) from anomalous transport (heavy, power-law or stretched tails), and the way it rescales with t exposes the self-similar, anomalous scaling form.

Non-Gaussianity. The departure of $P(\Delta r, t)$ from a Gaussian, summarized by the excess kurtosis (or the non-Gaussian parameter). A compact scalar that tracks how and when the displacement statistics become intermittent and heavy-tailed across scales.

Velocity autocorrelation $C(\tau)$. $C(\tau) = \langle v(t_0) \cdot v(t_0 + \tau) \rangle$, the correlation of the velocity with itself after a delay τ . It measures directional persistence and its memory: the decay time sets the ballistic-to-diffusive crossover, while a slow or sign-changing tail reveals persistence or anti-persistence.

Turning-angle distribution. The distribution of the angle θ between two successive horizontal displacement vectors along the trajectory – i.e. the change of heading from one step to the next. A distribution peaked at $\theta = 0$ means strong forward persistence (a correlated random walk); a flat distribution means memoryless reorientation. This is the microscopic origin of the persistence seen in $C(\tau)$ and the MSD.

First-passage and first-exit times. The time to first reach a given distance from the start, or to first leave a region of given size, as a function of that length scale. A scale-resolved probe of transport, often more sensitive than the MSD to heavy tails and to crossovers between regimes.

Speed and vertical-velocity distributions. The magnitude of the horizontal velocity and the vertical velocity. Basic kinematic characterization; the vertical velocity also separates climbing from sinking and feeds the segmentation below.

Trajectory geometry.

Tortuosity. How much a path winds, quantified by the ratio between the length actually travelled and the straight-line distance between its endpoints over the same interval (1 for a straight line, larger for a more convoluted path). It measures how far a flight departs from straight cruising and, as a single number, summarizes flight efficiency and the relative weight of winding versus directed motion.

Fractal dimension. The exponent describing how the apparent length of the trajectory grows as it is measured at finer resolution (estimated e.g. by box counting), ranging from 1 for a smooth straight line to 2 for a plane-filling, highly convoluted path. It captures the multi-scale convolutedness of the path in one index and is the geometric counterpart of the anomalous scaling seen in the MSD: directed cruising and area-restricted search occupy opposite ends of this scale.

Radius of gyration. The root-mean-square distance of the trajectory points from their centroid, i.e. the characteristic spatial extent of a flight. It sets the natural length scale of each flight and is used to normalize other observables so that flights of very different size can be compared on equal footing.

4.3 Segmentation into flight phases

Building on the previous work of Vilpellet et al. [4], each flight is segmented into the three phases of the soaring cycle:

Transition (glide). The inter-thermal cruise: a near-straight, energy-efficient glide toward the next source of lift (sinking, large directed horizontal displacement). This is the directed “step”.

Search. The exploratory phase entered when no thermal is found at a comfortable altitude: gentle turns and probing manoeuvres to sample the air mass and locate the next thermal while limiting altitude loss. It mediates the passage from one step to the next trap.

Climb (thermallng). Circling within rising air to regain altitude: persistent turning with small net horizontal displacement. This is the localized “trap”.

That work performs the segmentation with a hidden Markov model whose observations are a small set of *binary* trajectory-derived features — the sign of the vertical speed, the persistence of the curvature angle, and indicators of trajectory straightness.

A first task of this thesis is to scrutinize this segmentation rather than take it for granted: to assess whether those binary variables are adequate and robust on the larger, more heterogeneous dataset assembled here, and whether the feature set or the model should be refined, extended, or replaced. The segmentation method is therefore itself an object of study, and the specific variables are deliberately left open at this stage. This three-state description refines the two-state caricature underlying continuous-time random walks and Lévy walks.

4.4 Phase-resolved observables

The segmentation unlocks a second group of observables, defined per phase occurrence and then as ensemble distributions. These are the building blocks of an intermittent-transport description and, unlike those of Section 4.2, cannot be obtained without first segmenting the flight.

Climb. Duration τ_c (the *waiting time* of a trapping picture), altitude gained Δz , mean climb rate, and circling radius and period (the size and duration of one thermalling turn). The tail of the waiting-time distribution $\psi(\tau_c)$ controls whether trapping drives sub-diffusion.

Glide (transition). Horizontal step length ℓ (the *jump* of the walk), duration τ_g , ground speed, glide ratio (horizontal distance per unit altitude lost) and sink rate. The tail of the step-length distribution $\lambda(\ell)$ controls super-diffusive contributions.

Search. Duration τ_s , path length and tortuosity within the search, and the net displacement achieved. This quantifies the cost and efficiency of finding the next thermal – the phase in which prior work locates the signature of pilot skill.

Anatomy of a flight. Number of thermals (climbs) per flight, inter-thermal distances, and the fraction of both time and along-track distance spent in each phase: the coarse composition that sets the weight of each phase in the global transport.

Angular diffusion between glides. The turning angle between the heading of one transition segment and the next – the segment-level counterpart of the turning angle of Section 4.2. How fast successive glide directions decorrelate is the key ingredient of the minimal model below.

The quantities of central interest are the tails of $\psi(\tau_c)$ and $\lambda(\ell)$ (heavy, power-law tails being the signature of anomalous transport), the coupling between consecutive ℓ and τ (which discriminates a coupled Lévy walk from a decoupled CTRW), and the memory in the phase sequence (the transition matrix among the three states and duration–duration correlations).

4.5 Reproducing the reference results

A first objective of the analysis, undertaken once the segmentation is in place, is to check whether the enlarged—and eventually multi-source—dataset reproduces the quantitative and graphical results of Vilpellet et al. [4]. Two figures are the main targets.

The first is the per-phase conditional MSD (their Fig. 3): the transition segments should stay ballistic ($\alpha = 2$, i.e. $H = 1$) across time lags; the search phase should be strongly sub-diffusive ($H \approx 0.3$ for paragliders and hang gliders, ≈ 0.2 for sailplanes); and the climb phase should be quasi-ballistic at long lags but with an effective velocity about two orders of magnitude smaller, including the transient anti-correlation near the ~ 30 s thermalling period.

The second is the set of per-phase descriptive statistics (their Fig. 4): the glide ratios (≈ 8.5 , 10, and 25 for paragliders, hang gliders, and sailplanes), the horizontal-velocity statistics, the heavy-tailed distribution of transition times, the search-time fraction and the effective search radius, the vertical climb velocities, the thermalling radii, and the thin, near-exponential distribution of climb times—together with the global time budget ($\approx 60\%$ transition, $\approx 35\%$ climb, the remainder searching) and the global sub-ballistic scaling ($H \approx 0.88$).

Recovering these results would validate the acquisition pipeline and the segmentation and would test the universality of the reported behaviour on a substantially larger sample; systematic departures would be equally informative, pointing to dataset- or method-dependent effects to investigate. This comparison is therefore the bridge between the present data-acquisition stage and the original analysis that follows.

4.6 Transport analysis

Using these observables, test for anomalous transport: estimate the MSD scaling exponent α , fit the tails of $\psi(\tau)$ and $\lambda(\ell)$ with principled methods (e.g. maximum-likelihood power-law fits with goodness-of-fit tests), and perform model selection between the continuous-time random walk and the Lévy-walk frameworks [1, 2] – using the step/waiting-time coupling to tell them apart. Because logging devices, sites, gliders and weather differ, results will be stratified (by season, site, glider class, and data source) and trajectories normalized, and finite-size, censoring and sampling biases will be controlled throughout.

4.7 Modelling and simulation

The aim is a *minimal* stochastic model that reproduces the asymptotic anomalous scaling reported for soaring (a Hurst exponent $H \approx 0.88$) with as few ingredients as possible. A preliminary working hypothesis—to be made precise, and possibly revised, once the empirical phase statistics are available—is that the transport is carried mainly by the transition (glide) phases. The model would therefore resolve the transition segments explicitly while merging search and climb into a single effective non-transition state, so that the soaring cycle reduces to a sequence of directed glides interrupted by reorientations. Anomalous transport would then emerge from the directed transition segments combined with an *angular diffusion* of the heading from one transition to the next: successive glide directions decorrelate gradually rather than abruptly, producing a correlated random walk whose persistence controls the long-time exponent. Simulated ensembles would be compared with the data to test whether these ingredients suffice. These are baseline ideas, to be refined—or changed—as the analysis progresses.

4.8 Tooling

Mirror the acquisition package with an analysis package, and let the figures and tables of this document be generated automatically from the analysis outputs – as the dataset statistics already are – so that this document stays in step with the work and remains a reliable basis for the thesis and for presentations.

Bibliography

- [1] Ralf Metzler and Joseph Klafter. The random walk's guide to anomalous diffusion: a fractional dynamics approach. *Physics Reports*, 339(1):1–77, 2000.
- [2] V. Zaburdaev, S. Denisov, and J. Klafter. Lévy walks. *Reviews of Modern Physics*, 87(2):483–530, 2015.
- [3] Gautam Reddy, Antonio Celani, Terrence J. Sejnowski, and Massimo Vergassola. Learning to soar in turbulent environments. *Proceedings of the National Academy of Sciences*, 113(33):E4877–E4884, 2016.
- [4] Jérémie Vilpellet, Alexandre Darmon, and Michael Benzaquen. From random walks to thermal rides: Universal anomalous transport in soaring flights. *arXiv:2601.01293 [cond-mat.stat-mech]*, 2026.